

What remains to finish

PetSafe 100-1339 R03 A | v0.9.8 | 8 October 2026

All 150 catalog entries are drawn on one KiCad sheet. This includes empty options and test pads. The drawing is structurally complete; hidden wiring, candidate identities and component values still need evidence.

Drawing / libraries	Electrical gaps	Packages / values
1 A2 sheet; 20 stock + 4 authorized custom symbol definitions; no missing library files for selected candidates.	28 open pads = 10 populated-entry pads + 18 DNP pads. 0 GPIO pads open.	147/150 footprints; 44 capacitor values and 2 magnetic-part values unknown.

What your via review changed

User confirms RC2/V049-TP3 and RB0/V026-VREF. RC1/V053-V070-R18 confirmed; old RC1/VREF photo assumption explicitly rejected. Also confirmed: RA3-R14, RA4-R15, RA5-R16 and RC0-R11; V053-V071 rejected. 121 IDs reviewed; 120 active sites. PIC15/RC4 now reaches U5.2/INA via V022-V016; PIC16/RC5 reaches U5.3/INB via V023-V019. V011-V014/Q1.L now confirmed to motor VDD/U5 pin4, distinct from logic VDD. R43.2 joins TP14, separate from Q1.L. Six IDs remain unmentioned, some photo-matched. Motor VDD stays separate from regulated board VDD.

Remaining area	What is missing	How to resolve
PIC / block interfaces	0 GPIO pads open. Other pins have local nets but may lack remote continuations. Motor INA/INB now resolved to PIC15/16.	Review remote continuations of existing local nets; the open-GPIO checklist is complete.
Q8 / RF output / C6	V113/V115 resolve Q8 centre pin5 GND. V114 is U2 pin2 GND; Q8 source supply and C6.1 route remain inferred/unresolved.	V114 conflict resolved. Check Q8 supply independently; no inferred rail merge.
Tuning bank	All five controls mapped: RC6/R13, RA3/R14, RA4/R15, RA5/R16, RC0/R11. Capacitor-midpoint and antenna links remain unknown.	Seven capacitor pairs gave OL. Broad pairwise tests stopped; next trace toward a specific visible/component endpoint.
Auxiliary / receiver	V077 grounds C19/R29. V075 supply plus V064-V070 = 11.2k supports Q7 emitter/VDD, R19 pull-up and R18 base input; topology inferred. V067 joins J3.1/PIR supply; U6A remains DNP. V063 feed now joins regulated VDD; V017/R42.1/R44.1 now joins TP4/PIC25; Q1.L/V011-V014 motor rail confirmed; Q1 polarity/function unverified; R43.2/V015 joins TP14, separate from Q1.L.	RC1 now reaches R18 via V053-V070; no further generic GPIO sweep needed. V071/R21.2 reaches VREF/RB0; other receiver-bias branches and Q7 topology remain inferred. V053-V071 is rejected.
D2 / D6 / candidate parts	D2/4P identity and all three pad routes remain open. D6 has two open pads. V082 reaches logic VDD/TP103; its D3 terminal remains unresolved. U6/Q8 and small BJTs/diodes retain candidate identities/polarities.	D2 diode-mode guide; D6 continuity; targeted candidate pin checks.
Other open fitted pads	C26.1 and TP9.1; U6 NC pins 4/5 have unknown external ties. LED common now grounded from V127/photo; internal LED mapping and S1 legs remain provisional.	Targeted continuity; no need to identify chips on empty footprints.
Component values	44 unmarked fitted capacitor values and L1/L2 type/value remain unrecovered. C5 = 470uF / 16V is recorded; 38 fitted resistors have decoded nominal values.	LCR first on RF/tuning parts, recording frequency; in-circuit results are network readings.
Packages	147/150 footprints assigned. C5, S1 and LED1 still need exact geometry; other photo-based package assignments remain candidates.	C5 diameter/lead spacing; S1/LED1 body and pad spacing. R8/C11 sizes already recorded.

Remaining conflicts for later review

Resolved: V025, V026, V073 and Q8 V113/V115. **V114 resolved:** U2.T2/pin2 GND, not Q8. C5 negative confirmed by V125/V126. V127 grounds the photo-selected LED common pair. **V075:** regulated VDD measured. Q7/R19 terminal selection inferred from IMG_2434 and the new 11.2 kohm V064-V070 result. No functional test supplied.

Validation: KiCad 10.0.5 exports pass; 84 modeled nets match the native file. 36 ERC findings remain: 28 open pins, 3 isolated labels and 5 power findings. This checks the file, not hardware correctness. No artificial NC or power flags were added.

PIC connections: complete picture

Physical package pins, top view. Use the separate numbered close-up to locate them. OPEN means no modeled pad connection; a local connection can still have an unknown onward destination.

Pin	Function	State	Local destination / evidence
1	RE3/MCLR	Local net	J1 VPP; mapping inferred; D6 route missing
2	RA0	Local net	C25 signal pad / V031; onward destination unknown
3	RA1	Local net	R28 + R29 + TP7; user meter / visible trace
4	RA2	Local net	C39 signal pad / V035; onward destination unknown
5	RA3	Local net	V036-V108 to R14; user continuity report
6	RA4	Local net	V037-V102 to R15; user continuity report
7	RA5	Local net	V039-V097 to R16; user continuity report
8	VSS	Local net	GND / V041 photo anchor
9	RA7/OSC1	Local net	Y1 oscillator; photo-derived
10	RA6/OSC2	Local net	Y1 oscillator; photo-derived
11	RC0	Local net	V047-V091 to R11; user continuity report
12	RC1	Local net	R18.1 via V053-V070; user-confirmed; VREF link explicitly rejected
13	RC2	Local net	TP3 via V049; user-confirmed; existing R10 branch photo-derived
14	RC3	Local net	TP11; visible route; onward destination unknown
15	RC4	Local net	V022-V016 to U5 pin2 / INA; user-confirmed
16	RC5	Local net	V023-V019 to U5 pin3 / INB; user-confirmed
17	RC6	Local net	R13; user meter / visible trace
18	RC7	Local net	TP17; visible route; old R13 guess withdrawn
19	VSS	Local net	GND / C32 / V045 photo anchor
20	VDD	Local net	Regulated VDD / C32 / V042 photo anchor
21	RB0	Local net	VREF (TP104), V026 and V071/R21.2; user-confirmed
22	RB1	Local net	TP2 / R31; user meter / visible trace
23	RB2	Local net	TP1 / R30; user meter / visible trace
24	RB3	Local net	R4; photo-derived; sensing role remains inferred
25	RB4	Local net	TP4 / V017 / R42.1 / R44.1; user continuity; C41 photo-derived
26	RB5	Local net	TP10 / R35; user meter / visible trace
27	RB6/CLK	Local net	ICSP CLK / D4 / R32; photo-derived
28	RB7/DAT	Local net	ICSP DAT / D5 / R40; photo-derived

Every remaining open pad

Populated entries (10): C26.1, C6.1, D2.L, D2.R, D2.S, D6.1, D6.2, TP9.1, U6.L1, U6.L2.

Empty options (18): C42.1, C42.2, C49.1, C49.2, J6.1, J6.2, Q9.S, R36.1, R36.2, R46.1, R46.2, R47.1, R47.2, U6A.L, U6A.R2, U7.1, U7.2, U7.3.

Pad names above are photo-model names; U6.L1/L2 map to its internal NC pins 4/5. DNP pads do not block understanding of the fitted circuit. Vias with a component association do not automatically prove a connection to every other pad on a previously inferred net.

Evidence: user via review and supplied MX512H diagram; original front/rear photographs; reconstruction.json, via_audit.json, v09_net_changes.json, v091_net_changes.json, v092_net_changes.json, v093_net_changes.json, v094_net_changes.json, v095_net_changes.json, v096_net_changes.json, v097_net_changes.json, v098_net_changes.json and native KiCad validation. Pin names follow the existing Microchip datasheet audit. The via viewer retains reports and conflicts.